Durham-Jazan at AISA-ArabicFC: A Field-Wise QLoRA Ensemble and the Sensitivity of Exact-Match Scoring

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Abstract

We describe the Durham-Jazan submission to AISA-ArabicFC 2026, an Arabic function-calling task in which a system selects one tool from four candidates and fills its arguments, scored on tool choice (FnAcc) and on strict exact match of the whole argument dictionary (ArgEM). Across six QLoRA fine-tunes of ALLaM-7B and Qwen2.5-7B, trained on the provided data, negatives derived from it and 68 hand-written no-tool utterances, and run on one 16 GB GPU, we vote on each argument separately, then apply two deterministic rules estimated from training statistics. The system reaches ArgEM 0.824 on development under the final evaluator and 0.826 on the blind test, placing 9th of 17 in Track A, 8th of 17 in Track B and 11th of 21 in Track C. Our main findings concern the benchmark. The reference tool is listed first on every training and development row that calls one, in every data release. Shuffling the candidates at inference costs the voted system 0.202 FnAcc and 0.190 ArgEM, and retraining one adapter per base on permuted candidates removes the dependence with no loss of ArgEM. One post-processing rule contributes 0.028 ArgEM and is the only component whose gain separates from zero, the other targets a convention the final data release removed and is inert, and the largest effect we can measure comes from the benchmark's construction rather than from our modelling.

1 Introduction

Function calling underpins agentic language-model systems. It is the decision of whether a user request requires a tool call, which tool to invoke, and with what arguments. For English it is tracked by mature benchmarks such as BFCL (Yan et al., 2024), API-Bank (Li et al., 2023), and the suite released with ToolLLM (Qin et al., 2024). For Arabic, with its dialectal variation and diglossia, it has been measured far less often. The AISA-ArabicFC 2026 shared task (Nacar et al., 2026a), part of the wider AISA effort on agentic Arabic systems (Nacar et al., 2026b), addresses this with a single task over Modern Standard Arabic and four dialect groups. It is scored on tool selection (FnAcc) and on strict exact match of the predicted argument dictionary (ArgEM), with a reasoning-trace track (Track B) alongside the core track (Track A).

Team Durham-Jazan took part in both tracks. We fine-tune six LoRA (Hu et al., 2022) adapters with QLoRA (Dettmers et al., 2023) over two open 7B bases, ALLaM-7B (Bari et al., 2024) and Qwen2.5-7B (Qwen Team, 2024), combine them with a per-argument majority vote and a schema-derived enum-validity rule, and add two deterministic convention rules estimated from train-split statistics alone. Most of our remaining errors are failures of surface form rather than of comprehension, because the reference uses conventions the request underdetermines and, on the rows we fail, applies them inconsistently, so an answer right in substance can still score zero. We therefore measure the problem instead of modelling around it.

Shuffling the four candidates costs the system 0.202 FnAcc, since the reference tool is always listed first. We rank 9th of 17 in Track A, 8th of 17 in Track B and 11th of 21 in Track C. Our code, submitted predictions and data-use disclosure are released, with the adapters available on request.[1]

2 Background

Task setup. Figure 1 is one item end to end, from the labour-law domain. Each example gives an Arabic utterance, a context stating the current date and time, and four candidate schemas from a 27-tool registry over eight domains. The reference tool is not merely among the four. On all

[1]https://github.com/alshehriamal1994/aisa2026-durham-jazan

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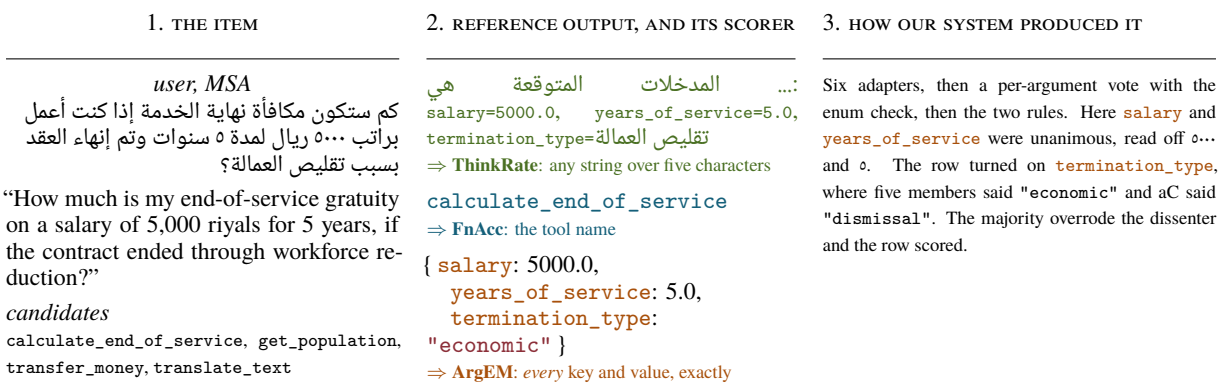


Figure 1: Development item 316, an end-of-service gratuity claim. Blue is the function name (FnAcc), orange the argument dictionary (ArgEM, all-or-nothing), green the trace (ThinkRate). The red value is the problem, and not because the mapping is wrong. The schema declares `Supported values: resignation, termination, retirement`, while the reference draws on eight values and uses one outside that list on 185 of 356 rows, with 300 of 491 for `calculate_zakat.type` and 38 of 40 for `search_quran.search_type`. A model obeying the schema it was given is scored wrong, which is why our enum-validity rule is inert.

11,000 training and development rows that call a tool it is the *first*, as the release stores them, in every release from v1.0 to v1.6 (Appendix E). The output is one tool with an argument dictionary, or `none`, plus an Arabic trace for Track B. FnAcc is exact-match tool selection over all rows. ArgEM is strict exact match of the whole normalised argument dictionary over rows whose reference calls a tool, not the full split. On development that denominator is 500 of the 545 rows, the other 45 calling no tool. On the blind split it is not published, so we cannot. The word normalised matters to our argument. The final normaliser forgives digits, orthography, list order and six closed value classes including currency, and since a July revision it also maps named days, relative terms and month names in date fields to one token, while an ISO date is compared exactly and is never reconciled with a natural-form date (Appendix C). ThinkRate is the share of rows carrying a trace, and the evaluator tests only that the string exceeds five characters, so despite the name it checks neither that the trace is Arabic nor that it relates to the call. Track A scores $0.4\,\text{FnAcc}+0.6\,\text{ArgEM}$ and Track B $0.3\,\text{FnAcc}+0.5\,\text{ArgEM}+0.2\,\text{ThinkRate}$, to which we sent the same files. Track C, published after the results and needing no separate submission, macro-averages per-dialect Track A scores and reports the max minus min *dialect gap*.

Data. 10,550 training, 545 development and 1,125 blind rows over MSA, Gulf, Egyptian, Levantine and Maghrebi. The blind split is MSA-heavy, 825 of 1,125 against 323 of 545. The data are template-generated and the reference is not consistent in the surface form it uses, a provenance that shapes Section 5.2. The reference was revised six times between 12 June and 23 July 2026, v1.1 to v1.6, and the scorer four times between 4 and 24 July. Section 4 states which pair each figure uses.

Related work. Function calling degrades outside English, the strongest model on MASSIVE-Agents falling from 57.4 to 36.1 AST accuracy between English and Arabic (Kulkarni et al., 2025). The closest Arabic-specific precedents are Ersoy et al. (2025), who study data strategies for Arabic tool calling and contribute training data and a tool-calling Fanar variant, and Kubrak et al. (2026), whose benchmark finds tool-calling accuracy dropping by five to ten points when models are prompted in Arabic. We pair ALLaM-7B, from the Arabic-focused line of Jais (Sengupta et al., 2023) and AceGPT (Huang et al., 2024), with the multilingual Qwen2.5-7B, reporting by dialect after NADI (Abdul-Mageed et al., 2024). Our analysis draws on the view that a single reference can misrepresent items with several acceptable realisations (Plank, 2022) and on what a benchmark score establishes (Bowman and Dahl, 2021; Sainz et al., 2023). Where Dehghani et al. (2021) show rankings moving with the choice of benchmark and Alzahrani et al. (2024) with small perturbations to its format, we measure what one benchmark's candidate order is worth.

3 System Overview

Fine-tunes. Six QLoRA adapters over 4-bit NF4 bases, three on ALLaM-7B and three on Qwen2.5-7B, with $\alpha = 2r$, dropout 0.05 and learning rate 2e-4 (Appendix A). The assistant target is `<think>`*trace*`</think>` followed by a JSON call (Appendix B), so Track B needs no separate procedure. We carry the trace of the member that won the tool vote, so on 24 rows it justifies a call whose values came from others. Reselecting traces to match the emitted dictionary would leave ThinkRate at 1.000, exactly as we scored, so the metric registers neither the defect nor its repair (Appendix C).

Negative augmentation. The train split is almost entirely positive and an unaugmented model almost never outputs `none` (FnAcc 0.932). We add candidate-swap negatives, real queries shown again with the gold tool removed from the candidate set, plus 68 no-action utterances we wrote by hand, each shown with six random candidate sets. FnAcc rises to 0.998 without costing a positive row.

Per-argument voting. The tool name is a majority over the six members. The argument dictionary is then decided *field by field*. Values are clustered by canonical form under the official normaliser, a key is kept when a majority holds it, and each kept key takes its most-supported value, with ties going to the earliest member in a fixed order. That order was fixed early and no longer tracks measured strength, though correcting it moves development ArgEM by one row, from 0.824 to 0.826. Combining structured output component by component is established in parser combination (Sagae and Lavie, 2006) and differs from self-consistency (Wang et al., 2023), which resamples one model. We have used cross-architecture majority voting before, for legal entailment (Alshehri et al., 2026), and the field-wise vote adapts it to structured arguments.

Field-wise combination is usually motivated by assembling an answer no member produced, which never happens here. Before the rules, the voted dictionary is identical to some member's on all 545 items, the gain coming from the 116 items on which the members disagree. The ensemble looks dispensable at $+0.002$ over the best single member, and that member with both rules reaches 0.822 against 0.824 for the vote, so with the rules applied the ensemble adds one row over its best member. A union oracle reaches 0.830, its 17-row headroom over the plain vote being argument rather than tool disagreements. An *enum-validity* rule then prefers a value appearing in the schema's `Supported values`, which changes no row's correctness on the six members we submitted.

Convention rules. Two deterministic rules, both estimated from the v1.4 train release alone and applied without reference to any development or test label. The *year rule* writes 2023 into a hotel or appointment date when the query states no year, since in that release the dated fields carried 2023 whatever the context date said, our models otherwise emitting context-consistent years that the metric penalised. Release v1.6 rewrote those fields to natural form without a year, so under the final evaluator the rule fires on 10 development rows and changes no score ($+0.000$, against $+0.004$ under the release we submitted against). The *omit rule* drops `recipient_iban` and `insurance_number` when the value is absent from the user text, since training keeps them only when stated ($+0.028$, alone or with the year rule). Appendix C gives the counts.

4 Experimental Setup

Data provenance. All six adapters were trained on the provided train split, negatives derived from it, and the 68 hand-written no-action utterances of Section 3, with no proprietary model, retrieval, external corpus or distillation in the submitted system, verified from training-log fingerprints in Appendix E. Only the inputs of the blind test were read and no development label entered training, though we did read the development negatives to decide which *categories* of no-action utterance to write.

Models and software. Transformers 5.5 (Wolf et al., 2020), PyTorch 2.6 for training and 2.10 for the blind run, PEFT 0.18, bitsandbytes 0.49, greedy decoding at batch 8. Scoring uses the official evaluator throughout, and local and leaderboard scores agreed exactly at every scored submission.

Evaluator provenance. Unless a figure is marked otherwise, every development number in this paper is scored with the final pinned evaluator: the leaderboard scorer at its last change, commit `30745ed0` of 24 July 2026, against data release

System	FnAcc	ArgEM	Ov.A	Ov.B
Random†	0.042	0.006	0.020	0.016
GPT-4o 3-shot†	0.868	0.699	0.767	–
GPT-4o 0-shot†	0.932	0.744	0.820	–
AISA-Think-270M†	0.981	0.679	0.800	0.807
Ours (dev)	0.998	0.824	0.894	0.911
Ours (test)	0.994	0.826	0.893	0.911
Best test, صقر (Saqr)	0.997	0.882	0.928	0.940

Table 1: Final official results. Ov.A and Ov.B are the Track A and Track B composite scores. †Official baselines on the blind test set (n=1,125), as re-scored by the organisers with the final evaluator for the overview paper. AISA-Think-270M is their fine-tuned FunctionGemma baseline (Nacar et al., 2026c), and GPT-4o produced no Track B trace. The pilot figures published on the portal during the campaign predate the normaliser and are not comparable. The top blind system is shown for reference. The development row is scored with the final evaluator (Section 4). Section 5.2 reports our submitted system at 0.796 FnAcc and 0.634 ArgEM once the candidate order is shuffled.

v1.6 of 23 July 2026, the pair the blind test was also scored with. The development leaderboard, and the version of this paper we submitted, used the scorer of 23 June and release v1.4. Appendix E gives the headline figures under both pairs and the released script that regenerates every development figure under either.

Compute. One RTX 4080 SUPER (16 GB) served all training and submitted inference, each fine-tune taking 2 to 5 hours and full blind inference about 1h50m, with the 32B comparison on one 80 GB GPU on Durham University's NCC cluster. The both-rules configuration was fixed as primary before the blind inputs were released (Appendices D and F).

5 Results

5.1 Official Results

Table 1 gives the official results and Table 3 what each component contributed. ArgEM moved from 0.824 to 0.826 across the two splits, so a system fixed before the blind data existed did not lose ground, and of the teams we can pair across the two splits on the development board's own figures, ours is the only one that rose, with the caveat that the board scored each development submission with the evaluator in force at the time and the blind test with the final one (Appendix D).

On Track C our per-dialect scores macro-average to 0.868 for 11th of 21, with a gap of 0.123 resting on a Maghrebi cell of 11 blind rows and not stable (Appendix G).

5.2 What the Scores Measure

Everything below is post hoc. The system of Section 3 was frozen before the blind inputs were released, and none of it altered our submitted predictions.

Competence against convention. On the 500 development positives the key set matches on 473 rows under the scorer's key handling, none of our 88 failing rows is a tool-selection error, and a date field is wrong on 29, most of them hotel stays where the v1.6 reference keeps the form the user wrote, *1 February*, and we emit a resolved ISO date (Appendix H).

Forgiving surface form on dates and currencies gives a *convention-relaxed* 0.884 against 0.824 strict, neither official, so convention does not explain the whole distance (Appendix H).

The ordering carries the tool decision. Keeping our system's decision of whether to call a tool but replacing its choice of which with "always take the first candidate" leaves development FnAcc at 0.998, success without reading the request being the signature of an annotation artefact (Gururangan et al., 2018). Shuffling the candidates and re-running inference settles which cue our models used. FnAcc falls to 0.796 and ArgEM to 0.634, drops of 0.202 and 0.190, against the 0.323 a positional system scores. The dependence is uneven. Our strongest member loses 0.064 and answers first on 28.3 per cent of rows, against the 26.2 per cent where the reference sits. Our rank-64 ALLaM member loses 0.650, answers first on 90.5 per cent, and lands on the positional baseline. Models carry option-order sensitivity even without fine-tuning (Pezeshkpour and Hruschka, 2024; Zheng et al., 2024). Ours saw that order in training, and retraining one adapter per base with the order permuted removes the drop at no cost in ArgEM (Appendix H), so the sensitivity measures what the benchmark taught rather than a defect of the models. FnAcc carries 0.4 of Track A, and for a system that reads the position, as ours did to varying degrees, a substantial share of that weight rewards the ordering rather than the ability it names.

6 Conclusion

We presented a field-wise vote over six QLoRA fine-tunes with two train-derived rules, which placed 9th of 17 in Track A. One train-derived rule contributes more than the ensemble, the other became inert when the final data release removed the convention it encoded, and the fixed candidate order substantially inflates FnAcc, confounding tool selection with a positional shortcut. Shuffling the four candidates before release would close that shortcut. For the two adapters we retrained, one per base, it cost nothing in ArgEM, and whether that holds for other systems is untested.

Limitations

Our headline deltas are small against the size of the splits. On 500 development positives one row is 0.002, and only the omit rule and the two rules together separate from zero under a paired test (Table 3). The ensemble's advantage over its best single member does not, and that comparison is in any case against a member chosen on the same split. The blind figure of 0.826 rests on a positive count the organisers do not publish. Working back from the published score gives roughly a thousand rows, on which the binomial standard error is about 0.012, so the 0.002 difference from our development score should be read as the absence of a fall rather than as a measured gain.

Our analysis of other teams is weaker still. We pair four systems across the two splits using published leaderboard figures, we did not see their systems, and we cannot attribute their regression to development fitting rather than to anything else. The direction is consistent across all four, and that is all we claim.

Both rules were estimated from the v1.4 train release. Evaluated against the final release the year rule is already inert, which is the transfer failure this caveat anticipated, and the omit rule's value may likewise not survive a later revision.

The convention-relaxed score of 0.884 is our own construction, not an official metric. It forgives surface form on currency and date fields as a class, which is coarser than forgiving only the cases where the reference is demonstrably inconsistent, so it is an upper estimate of what convention costs us. It is also coarser than the official normaliser, which aliases qualified currency names to ISO codes and, since 23 July, the bare ريال, درهم, جنيه and دولار to SAR, AED, EGP and USD, resolving a genuine ambiguity in favour of one reading, so a Moroccan درهم against a reference of MAD still fails (Appendix I).

We did not train on development labels, but we did read the development negatives to decide which categories of no-action utterance to write by hand. Our development function accuracy is therefore mildly optimistic, and only the blind figures are free of that.

Our candidate-order result has its own limits. That the reference tool is first on every row is a fact about the release, and the shuffled run shows our own members depended on it to different degrees. It does not show that other teams' systems did, since we have neither their models nor their predictions, and we do not claim it. We retrained only one adapter per base, not all six members, and the result speaks only for our own models.

Finally, the dialect gap is fragile on these data. Maghrebi is 11 of the 1,125 blind rows, so a single row moves that cell by about nine points, and any max-minus-min statistic over five cells inherits that noise.

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A Hyperparameters and Software

Bases: `ALLaM-AI/ALLaM-7B-Instruct-preview` and `Qwen/Qwen2.5-7B-Instruct`, 4-bit NF4 with double quantisation, bf16 compute. Members are listed in Table 2. LoRA on `q,k,v,o,gate,up,down_proj`, dropout 0.05, learning rate 2e-4, cosine schedule, warmup 0.03, `paged_adamw_8bit`, maximum sequence length 1024, per-device batch 1 with gradient accumulation 32 for the two members whose training command is preserved (`aC` and `qC`). The exact accumulation used for the four earlier members was not recorded, and we say so rather than assume it matched. Its practical effect is bounded by the spread between members, which is 0.020 ArgEM end to end, ten rows of 500, with the four unrecorded members spanning 0.774 to 0.790 inside it. The configuration was adopted from common QLoRA practice rather than searched on this task, with rank and epochs varied across members (Table 2) rather than tuned per member. Inference: greedy, `max_new_tokens` 256, batch 8. Adapters are LoRA deltas of about 80M trainable parameters, and 160M for the rank 64 member.

B Prompt Format

Training and inference build the input identically. The system prompt is in Arabic and carries an instruction block, the four candidate schemas serialised as JSON, and a line stating the context date. The user message is the utterance verbatim. The assistant target is `<think>`*trace*`</think>` followed by a single JSON line of the form `{"name": t, "arguments": {...}}`, with `{"name": "none", "arguments": {}}` when no tool applies. Argument values are serialised with `ensure_ascii=False` so the model imitates the reference's exact rendering, floats as `5000.0` and

Arabic strings unescaped. The exact template is `src/aisa/prompt.py` in the released repository.

C The Combiner

Let M be the six members and canon$(\cdot)$ the official normaliser. For one item:

1. Each $m \in M$ emits a tool t_m and a dictionary A_m.
2. $t^\star$ is the tool with most votes among $\{t_m\}$, ties broken by the fixed member order.
3. Let $M^\star = \{m : t_m = t^\star\}$. A key k is kept if it appears in A_m for at least half of $M^\star$, so with six agreeing members a three-against-three split on a key keeps it.
4. For each kept k, group the values $\{A_m[k] : m \in M^\star, k \in A_m\}$ by canon, take the largest group, and emit that group's value from the earliest member in the fixed order. Ties between groups are broken the same way.
5. If a schema lists `Supported values` for k, restrict to the groups whose value is a member of that list and take the largest of those, if any.
6. Apply the year rule, then the omit rule.

The final normaliser folds Arabic-Indic and ASCII digits, strips diacritics and normalises alef and ta-marbuta, equates number words with numerals, compares list values as sets, strips optional arguments per tool, aliases six closed value classes (country, city, currency, language, zakat type and search type) together with product names, specialties and categories, and in date fields maps named days, relative terms and month names across Arabic and English to one token, so الجمعة and `Friday` are one string to the scorer. An ISO date is compared exactly and a natural-form date is never reconciled with a resolved one, so `2023-02-01` and `1 فبراير` stay distinct. Bare ريال, درهم, جنيه and دولار have been aliased to SAR, AED, EGP and USD since 23 July 2026, while bare دينار and ليرة are not. The scorer we submitted against, of 23 June 2026, had none of the date handling and none of the bare-currency aliases.

The members agree outright on 429 of the 545 development items. The enum-validity rule was worth $+0.002$ on an earlier seven-member ensemble and we keep it only because it cannot prefer a value the schema forbids.

Because the trace is carried from the member that won the tool vote, it can contradict the emitted dictionary. Reselecting it from a member whose dictionary matches the vote removes the mismatch on all but the 21 rows where the convention rules leave a dictionary no member held, so the defect is structural to post-editing rather than to voting alone. Removing it entirely would require the trace and the call to be decoded together against the final dictionary rather than assembled from separate members, which a field-wise combiner cannot do by construction. We found this after the blind phase closed.

The year rule rests on the v1.4 train release, where 666 of the 816 ISO-formatted `search_hotels` date fields carried 2023, as did all 48 ISO-formatted `book_doctor_appointment` dates. Release v1.6 rewrote those fields to natural form: of 1,095 non-empty hotel date fields only 76 still carry a year, 70 of them 2023, and 6 of 520 appointment dates, all 2023, so on the final release the rule is inert (Table 3). The omit rule rests on `recipient_iban` being kept 278 times and `insurance_number` 397 times. Once Arabic-Indic digits are folded to ASCII, the value is present in the user text in every case and there are no counter-examples. Our implementation compares raw strings without folding digits. On development that alters the fired set on one row and changes no score, though folding is the safer implementation and we would adopt it.

The year rule fires on `search_hotels.check_in`, `search_hotels.check_out` and `book_doctor_appointment.date`. It rewrites a predicted year to 2023 when that year is not already 2023 and does not appear in the user text. The omit rule deletes `recipient_iban` or `insurance_number` when the predicted value is not a substring of the user text. Both tests are plain substring matches on the raw strings.

D Blind-Phase Protocol

Following van Miltenburg et al. (2021) we fixed the both-rules configuration as primary before the blind inputs were released, using the remaining slots for the rule ablation of Appendix F rather than for further tuning. Development-fitted gains did not transfer for the four teams we can pair across the two splits. All four fell while ours rose from 0.824 to 0.826, and the two furthest above the eventual blind range fell furthest, 0.932 to 0.869 and 0.886 to 0.708. With four paired systems and no sight of their methods this is a direction rather than

tag	base	r	ep	release	ArgEM	order drop
v10	ALLaM-7B	32	2	v1.1	0.782	−0.233
v12	ALLaM-7B	64	2	v1.1	0.776	−0.650
v7b	Qwen2.5-7B	32	2	v1.1	0.790	−0.154
v7	Qwen2.5-7B	32	1	v1.1	0.774	−0.328
aC	ALLaM-7B	32	1	v1.3	0.774	−0.160
qC	Qwen2.5-7B	32	1	v1.3	0.794	−0.064

Table 2: The six members. $\alpha = 2r$ throughout. Members differ by base, rank, epochs and data release. We did not vary the seed. ArgEM is measured on the 500 development positives under the final evaluator (Section 4), and order drop is the member's FnAcc loss when the four candidates are shuffled at inference (Appendix H).

an effect estimate. It is also confounded by the evaluator: the development board scored each submission with the scorer and release in force when it was made, and the blind test used the final pair, so part of any movement may be the evaluator rather than the system. For our own system the two evaluators differ by 0.002 on development. The comparison therefore says something about our system and nothing firm about the others.

E Reproduction and Provenance

`bash scripts/run_blind.sh <test.jsonl> <out.jsonl>` regenerates the blind submission deterministically from the six adapters. The same command on the development split reproduced our submitted development scores exactly under the scorer of 23 June with release v1.4, the pair the development leaderboard used: ArgEM 0.822 / Overall A 0.8925 / Overall B 0.9104. The same development file scores 0.824 / 0.8937 / 0.9114 under the final evaluator. Under that earlier pair the shuffled-order run scored 0.636 ArgEM, a drop of 0.186 over 96 rows lost and 3 gained, which the submitted version misreported as 0.184. Between the two pairs, 22 rows we had right become wrong, every one through the v1.6 relabelling of hotel dates and destination countries, and 23 rows we had wrong become right, 21 of them through the scorer's date and currency aliases. `scripts/rescore_final_evaluator.py` in the released repository regenerates every development figure in this paper under either pair from the released prediction files, and `scripts/audit_candidate_order.py` checks the position of the reference tool on any release of the dataset, as the release stores the candidate schemas and as they are rendered into the prompt. The reference is first on 10,500 of 10,500 training and 500 of 500 development positives in all seven releases from v1.0 to v1.6. Tokenised-count fingerprints from the training logs (11,284 for the v11-era members, 12,369 for the v1.3-clean members) uniquely identify the training file of every member as the provided train split plus the derived and hand-written negatives. The parent corpus, the larger collection the shared-task splits were drawn from, contributed nothing to the submitted system, verified by rebuilding the combiner's tool registry three ways (from the parent, from the shared-task train split, and from the blind rows' own schemas) and finding all three byte-identical.

Leakage audit. Of the 500 development positives, 26 occur verbatim in the train split and 292 have a train twin at or above 0.8 character similarity, candidates retrieved by character n-gram tf-idf and the top ten rescored exactly. On its own that figure says little. Applying the identical procedure to 500 training queries against the rest of the training split, each query left out of its own comparison, gives 304 twins at the same threshold and a median best-twin similarity of 0.820 against 0.816 for development. The development split is therefore no closer to the training split than the training split is to itself. The corpus is redundant by construction and the splits are not differentially leaky, so the informative quantity is the transfer rather than the overlap. A 1-nearest-neighbour predictor over all 10.5k train rows scores only 0.40 per argument and 0.23 exact match under the evaluator of the submitted version, because the twin carries re-randomised fills. Near-duplicate overlap is known to inflate scores where the answers carry over (Lewis et al., 2021), which here they do not, and n-gram decontamination would call these splits clean (Yang et al., 2023). On the blind split 5.2 per cent of queries are verbatim in train, matching development before de-duplication.

Both split-comparison diagnostics are reproduced by `scripts/blind_determinacy.py` in the released repository, and both run on model outputs and inputs alone, so neither requires the blind gold labels. Unanimity is computed over the 1,003 blind rows on which a majority of *our members* call a tool, against the 500 gold positives on development. These are not the same quantity, since the blind positives are not published, so the comparison is approximate. The submitted file con-

tains 1,004 calls, the extra row being one where the members split three against three. Twin similarity uses the same difflib character ratio as the original audit, retrieving candidates by character n-gram tf-idf and scoring the top ten exactly.

F Blind Submission Variants

Development, n=500	ArgEM	95% CI	p
best member (qC)	0.794		
weakest (v7, aC)	0.774		
vote + enum	0.796	$[-.014, +.018]$	1.00
+ year rule	0.796	$[.000, .000]$	1.00
+ omit rule	0.824	$[+.014, +.042]$	$<.001$
+ both (V1)	**0.824**	$[+.014, +.042]$	$<.001$

Table 3: What each component contributed, on the 500 development positives under the final evaluator. Intervals are 2,000-sample paired bootstrap and p values are exact McNemar. The vote row is tested against the best single member, and each rule row against the vote. The ordering is the point. The omit rule is worth $+0.028$ and is the only component that separates from zero. The year rule changes the outcome of no row under release v1.6 (it was $+0.004$ under v1.4), and the ensemble's $+0.002$ over the best single member does not separate from zero.

We prepared the four rule variants of Table 3. The task permitted three submissions per track, of which only the best-scoring file is ranked. V1 (both rules) was pre-registered as primary and submitted to both tracks, with the remaining slots used for the other variants. All four files carry 1,125 rows, ThinkRate 1.0, no out-of-candidate tools and 121 `none` rows. On the blind inputs the year rule fired on 17 rows (all `search_hotels`) and the omit rule on 36 rows, and the pairwise differences between the four files range from 17 to 53 rows. The remaining headroom is of the same kind as the rules themselves. Our models emit `termination_type` on 8 development rows where the reference omits the field, so a third rule of the same family would be worth $+0.016$, eight times what the ensemble contributes.

G Track C in Detail

The gap is not a stable quantity. Bootstrapped development intervals for every non-MSA cell overlap the MSA interval, and a max minus min over five cells is dominated by the smallest.

Development ArgEM by dialect with 95% bootstrap intervals over 2,000 resamples: MSA $0.855\,[0.813, 0.896]$ on 289 rows, Gulf $0.833\,[0.750, 0.917]$ on 72, Egyptian $0.742\,[0.636, 0.848]$ on 66, Levantine $0.776\,[0.672, 0.866]$ on 67, and Maghrebi $0.667\,[0.333, 1.000]$ on 6. No non-MSA interval is disjoint from the MSA interval.

Under the final evaluator the hotel date pair leads the failing fields in every dialect. Beneath it Levantine keeps `country`, Egyptian list-valued `items`, and MSA `termination_type`. On the blind split the organisers' per-dialect Overall scores are 0.911 MSA, 0.901 Gulf, 0.883 Levantine, 0.856 Egyptian and 0.788 Maghrebi, macro-averaging 0.868 for 11th of 21 with a gap of 0.123. These are the organisers' published figures. They cannot be recombined into the Track A row of Table 1 without the per-dialect row and positive counts, since FnAcc is computed over all rows and ArgEM over positive rows only, and those counts are not published. Balance looked like our clearest edge on development, where our ArgEM gap of 0.080 was the smallest on that board as the evaluator then in force scored it, and it only partly survived. Under the final evaluator the same development gap is 0.188, the six-row Maghrebi cell having lost one row. Our blind gap is narrower than the 0.136 and 0.140 of the two highest-scoring systems but wider than five others, and the system with the smallest gap of all placed third in Track A. Balance and accuracy were not in tension, and our narrow development gap was not the advantage we had taken it for. Maghrebi is 11 of the 1,125 blind rows, so one row moves that cell nine points.

H Further Analyses and Negative Results

All figures below are measured on the development set against V1, the configuration we fixed before the blind phase. The diagnostics come first, and the closing paragraph reports approaches that did not help.

Candidate order. Re-running inference on the development split with the four candidates shuffled, everything else held fixed, costs the voted system 0.202 FnAcc and 0.190 ArgEM. Per member the loss runs -0.064 (`qC`), -0.154 (`v7b`), -0.160 (`aC`), -0.233 (`v10`), -0.328 (`v7`) and -0.650 (`v12`). The last answers with the first candidate on 90.5 per cent of rows, against the 26.2 per cent where the reference sits, and its shuffled FnAcc of 0.349 lands close to the 0.323 a purely positional system scores. Retraining one adapter per base on the same data with the candidate order permuted

(seed 20260816, all else per the released recipe) removes the dependence. Shuffled-order FnAcc rises from 0.839 to 0.985 for ALLaM and from 0.936 to 0.996 for Qwen, at a canonical-order cost of 0.017 and 0.006, with no loss of ArgEM, and both retrained adapters answer with the first candidate at the rate the reference sits there. The dependence is therefore a property of how the release orders its candidates rather than of the models, and shuffling before release would remove it. Grouped by base among the submitted members, ALLaM appears twice as exposed as Qwen, -0.347 against -0.182, but that gap is carried entirely by `v12`, which is also the only member at rank 64, so base and rank are confounded, and three members per base are too few to separate them. Excluding it the two bases are indistinguishable at -0.196 and -0.182. The system loses 97 ArgEM rows and gains 2. Of the 97, 88 are rows whose tool flipped and 9 are rows that kept the tool but lost an argument, so the effect falls predominantly on tool selection.

Where the residual sits. All six members are wrong on 85 of the 500 positives, and 71 of our 88 errors fall on such rows. An oracle allowed to pick the best member on every row would reach 0.830 against 0.824 for the vote, so most of what we get wrong is beyond any combiner. None of the 88 is a tool-selection error. Counting fields rather than rows, a field failing when it is missing on one side or its values do not match under the normaliser, the date family gives 54 instances over 29 rows, `check_in` and `check_out` 25 each and always on the same hotel row, and `date` 4. The rest is a thin tail: `termination_type` 9, of which 8 are a key the reference deleted, `destination_country` 8, all a key the reference deleted, `items` 7, `procedure` 6, `country` 6, and `medication_name` 5.

Surface form. In release v1.6 the `date` field alone takes 111 surface values in three scripts, which the final normaliser reduces to 59, and the whole date family 427, reduced to 309. What the normaliser does not do is reconcile a natural-form date with a resolved one, and resolving both sides to a calendar date before comparison, as Ersoy et al. (2025) do, would remove our largest class of errors. Worked examples are in Appendix I.

Scale. A Qwen2.5-32B QLoRA reaches ArgEM 0.786, against 0.824 for the six-member vote and 0.794 for the best single member. Family and recipe were held fixed, which is what makes it a controlled comparison, but it also means the rank, learning rate and batch size were those chosen for a 7B model and nothing was searched at 32B. We report it as one point under our own recipe and draw no conclusion about scale on this task, beyond noting that exact-match metrics penalise semantically correct output differing in surface form (Zhong et al., 2020). The official baselines, rescored by the organisers with the final evaluator, put prompted GPT-4o at 0.744 ArgEM zero-shot above the fine-tuned FunctionGemma-270M (Gemma Team, 2025; Nacar et al., 2026c) at 0.679 on the blind test, the reverse of the order the pilot figures on the portal had shown, and that pair confounds tuning with scale in any case.

Nine further approaches. The deltas in this paragraph were measured while the system was being built, under the scorer and release then current, v1.1 to v1.4, so they are comparable with each other but not with the figures above. A canonicaliser learned from train, which changes no development row, the residue being plausible variation rather than correctable error (Plank, 2022). Dialectal paraphrase augmentation -0.014. Ensemble self-distillation, no gain over the vote and about 26 hours to label. Canonical-output supervision -0.05. Cross-lingual co-training -0.014. Combiner weight tuning, where the in-sample gain does not survive held-out folds. A train-pool kNN argument fallback, negative because the fills are re-randomised. Retrieval-augmented few-shot prompting, 0.796 to 0.742. Adding the parent corpus the splits were drawn from, -0.046, since its label conventions differ from the shared task's. A learned verifier over member disagreements, where the recoverable cases are dominated by contradictory gold so no verifier wins both sides.

I Qualitative Examples

All examples are development rows. Translations are ours. Our only development tool-selection error falls on a negative row, so it is not among the 88 argument failures above. It is item 509, ليش الجو في الخليج حار أكثر من لبنان؟ ("Why is the weather in the Gulf hotter than in Lebanon?"), where the reference is `none`, three of the six members call `get_weather` anyway, and the resulting three-against-three tie is resolved in their favour by the member order.

Canonicalisation, learned. For ممكن تترجم ده للألماني؟ أنا مشغول جدا النهاردة (Egyptian: "Can you translate this to German? I am very busy today"), the system predicts `translate_text{target_language: "de", text: "أنا مشغول جدا النهاردة"}`, correctly ISO-coding the language and extracting the span to be translated rather than the request boilerplate.

The convention tax, illustrated. These are gold and prediction pairs from the 30 convention-tax cases under the final evaluator, in which both forms are valid. Twenty-five are hotel stays where the reference keeps the form the user wrote, as in من 1 فبراير إلى 10 فبراير ("from 1 February to 10 February"), against which our resolved `2023-02-01` fails because an ISO date is compared exactly. The direction flips between rows: for يوم الأربعاء ("on Wednesday") the reference is the resolved `2023-10-11` and our الأربعاء fails. The remainder are forms outside the alias table, نهار الجمعة against الجمعة for Friday, the Egyptian spelling الحد الجاي against الأحد الجاي for next Sunday, and a Moroccan درهم against a reference of `MAD`, which the normaliser reads as AED. Under the scorer we submitted against, الجمعة against `Friday` and الأسبوع الجاي against `next week` were also among the cases, and the year rule targeted references such as `2023-02-01` against our `2024-02-01`, a generator stamp the v1.6 release removed.